Multimodal Heterogeneous Hypergraph Learning for Incomplete Multimodal Semantic Segmentation of Remote Sensing Images

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***Abstract*—Advancements in multimodal remote sensing have enhanced image interpretation by enabling the integration of complementary information from heterogeneous sensors. However, in real-world scenarios, modality availability is often inconsistent due to sensor limitations and environmental factors, leading to incomplete data across regions. In this article, we propose a novel Multimodal Heterogeneous Hypergraph Learning (MHHL) approach for incomplete semantic segmentation of remote sensing images. The proposed MHHL framework constructs a heterogeneous hypergraph model to represent the complex relationships among modality combinations under incomplete conditions. It further establishes a graph neural network-driven transductive learning framework to enable mutual learning and feature refinement across different modality combinations. The framework performs high-order relational modeling through a hypergraph and utilizes the representation aggregation capabilities of graph neural networks to dynamically update hypernode features, thereby constructing a comprehensive multimodal feature space and effectively aligning feature distributions across incomplete modalities. Additionally, the framework introduces global multimodal feature modeling and a multimodal hybrid feature discriminator to address the feature shift problem caused by varying modality combinations. Experimental results demonstrate that our MHHL model significantly outperforms other relevant deep learning networks in terms of accuracy for incomplete multimodal semantic segmentation.**

***Index Terms*—Incomplete multimodal learning, multimodal fusion, remote sensing, semantic segmentation.**

I. Introduction

REMOTE sensing imagery has become an essential tool across various fields [1], including urban planning and environmental monitoring [2], offering detailed insight into the Earth's surface conditions [3]. With the continuous progress in sensor technology, there has been a significant increase in the availability of diverse data types, including optical, multispectral, Synthetic Aperture Radar (SAR), and Light Detection and Ranging (LiDAR) [4], capturing the same geographical region. These diverse datasets provide complementary views, enhancing the interpretation of remote sensing images for more accurate decision-making [5]. Furthermore, the rapid development of deep learning techniques has greatly improved the capacity to automatically analyze these heterogeneous datasets, pushing the boundaries of automated and precise remote sensing analysis [6].

Pixel-level interpretation of multimodal remote sensing images is typically approached through two methods: pixel-wise land cover classification [7] and semantic segmentation [8]. Both aim to assign each pixel to a semantic category but differ in their methodologies [9]. Pixel-wise classification treats each pixel independently, using its feature vector and the surrounding neighborhood's features [10]. For example, Li et al. [11] proposed a spectral correction module to enhance classification accuracy by reducing noise between hyperspectral and LiDAR images. On the other hand, semantic segmentation integrates information by dividing the image into coherent regions representing different semantic classes. This method is particularly useful for large-scale tasks like land cover mapping and urban analysis. Li et al. [12] developed the WHU dataset and a multimodal cross-attention network to effectively fuse optical and SAR images, improving classification accuracy for land use. These approaches demonstrate that multimodal fusion enhances classification accuracy by addressing the limitations of individual data sources.

However, most existing methods assume the availability of all modalities during both training and inference. This assumption is often impractical, as real-world applications frequently encounter incomplete modality inputs due to sensor coverage gaps, occlusions, or adverse weather conditions. As illustrated in Fig. 1, some regions may contain complete multimodal data, while others lack one or more modalities. These incomplete scenarios pose significant challenges for semantic segmentation, often leading to degraded performance or complete failure in missing-modality regions. Moreover, many current approaches rely on pairwise fusion or sequential cross-modal interactions, which may suffice under fully paired settings but are inadequate for capturing high-order, nonpairwise dependencies inherent in multimodal data [13]. This limitation becomes more pronounced under partial modality availability, where such models struggle to capture the complex intermodal semantics and exhibit poor generalization to heterogeneous or dynamic environments. In contrast, hypergraphs provide a principled framework for modeling multiway relationships among modality combinations. By connecting multiple nodes within a single hyperedge, they enable high-order

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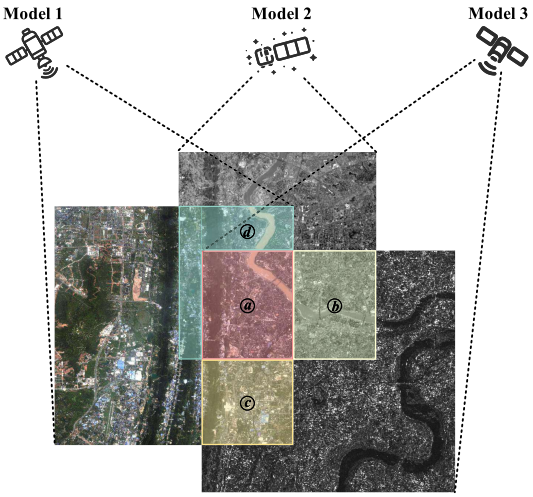


Fig. 1. Illustration of incomplete multimodal data coverage across different regions by multiple satellites. Region ⓐ represents the area covered by all three satellites, while regions ⓑ, ⓒ, and ⓓ depict areas covered by two satellites. The remaining regions are covered by only one satellite, highlighting the incomplete nature of data coverage across these areas.

interaction modeling irrespective of modality presence. This makes hypergraph-based modeling particularly well-suited for robust semantic segmentation under incomplete multimodal conditions.

The core challenge in incomplete multimodal fusion for remote sensing lies in effectively learning and integrating features from partial modality combinations to ensure reliable segmentation performance under diverse and complex real-world conditions. To address this problem, this article proposes a heterogeneous hypergraph learning framework for incomplete multimodal semantic segmentation of remote sensing. In this framework, a heterogeneous hypergraph is constructed to capture the complex relationships among diverse modality combinations, where each modality configuration is represented as a hypernode, and connections are established based on shared modalities. This structure facilitates cross-combination information propagation, enabling the model to compensate for missing modalities by leveraging knowledge from overlapping modality subsets. A transductive graph learning strategy is employed to iteratively update hypernode features, thereby constructing a unified and robust multimodal feature space. Furthermore, to mitigate feature distribution shifts caused by modality incompleteness, a multimodal hybrid feature discriminator is introduced to align feature representations across different modality configurations. The overall architecture of the proposed framework is illustrated in Fig. 2. The main contributions of our work are as follows.

1) We propose a multimodal heterogeneous hypergraph model that employs diverse heterogeneous hypernodes to represent various modality combinations under incomplete modalities, with hyperedges capturing the relationships between these incomplete modalities. The heterogeneous hypergraph model effectively facilitates the representation and association of multimodal features across different combinations, enabling incomplete modalities to learn comprehensive fused features from complete modalities.
2) We propose a transductive learning framework based on graph neural networks to facilitate mutual learning among different modality combinations. Leveraging the hypergraph structure and graph neural networks, the framework dynamically refines the hypernode features of various modality combinations, fostering a unified and enriched multimodal feature space.
3) We propose a multimodal hybrid feature discriminator to tackle the feature shift problem arising from varying modality combinations. By modeling global multimodal feature distributions, the proposed method achieves consistent feature alignment, contributing to improved robustness and accuracy in semantic multimodal segmentation.

The article is organized as follows: Section II reviews current approaches to incomplete multimodal semantic segmentation. Section III details the proposed deep network for incomplete multimodal segmentation. Section IV presents the experimental results and analyses, while Section V concludes the study.

II. Related Work

A. Multimodal Semantic Segmentation for Remote Sensing

Semantic segmentation of remote sensing images plays a critical role in applications such as urban planning [14], land cover analysis, and crop monitoring. Recent developments in deep learning have significantly propelled advancements in this field. Attention-based deep learning models, particularly Transformer-based architectures, have gained prominence for their ability to capture global context and handle complex object boundaries. For example, ST-UNet [15] integrates the Swin Transformer into the traditional U-Net framework, achieving improved segmentation accuracy through enhanced global context modeling and spatial interaction.

In addition to advancements in model architectures, multimodal fusion has become a critical area of research to improve remote sensing image segmentation [16]. The proliferation of multisource remote sensing data, including optical, hyperspectral, SAR, and LiDAR, has driven the development of multimodal fusion techniques. These techniques leverage the complementary strengths of different modalities to produce more comprehensive and accurate segmentation results [17]. Xu et al. [18] proposed a dual-branch dynamic modulation network for hyperspectral and LiDAR data fusion, which leverages dynamic gradient optimization, bidirectional feature enhancement, and feature distribution consistency loss to reduce intermodal interference and improve classification performance. The CMX [19] framework is a unified multimodal fusion approach that enhances RGB-X image semantic segmentation by employing cross-modal feature rectification and long-range context exchange, thus achieving effective fusion and generalization across various sensor data. Similarly, DFormer [20] is a novel RGB-D pretraining framework that learns transferable representations for segmentation tasks by pretraining with image-depth pairs and utilizing a sequence

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of specialized RGB-D blocks. The aforementioned three frameworks focus on feature fusion methods, where feature fusion aims to merge multimodal features within the feature space to derive more comprehensive and complete representations.

In summary, advancements in deep learning and multimodal fusion have significantly improved semantic segmentation in remote sensing, particularly by integrating diverse data sources to enhance accuracy. However, real-world applications often face the challenge of incomplete modalities, which can severely affect the performance of feature fusion methods. Therefore, it is crucial to develop robust methods for multimodal semantic segmentation that can effectively deal with incomplete modalities, ensuring reliable performance even under conditions of incomplete modalities.

B. Incomplete Multimodal Learning

Multimodal sensors have become indispensable across various domains, making the ability to handle missing modalities a critical concern for real-world applications. In fields like healthcare, for instance, patients may only have access to certain imaging scans, such as MRI or CT, leading to a heightened focus on incomplete multimodal learning. Approaches in this area often revolve around either synthesizing the missing modalities [21] or learning a shared latent space that accommodates all modalities [22].

In the realm of computer vision, several notable methods have been proposed to address the challenges of incomplete modalities. Wang et al. [23] introduced a distribution-consistent modal recovery approach, which maintains distribution consistency by transferring information from the available modalities to the missing ones. Lee et al. [24] proposed a multimodal prompt learning framework that specifically targets the difficulties posed by missing modalities during both training and testing. SMIL [25] tackles the issue of missing modalities and feature space regularization using a feature reconstruction network, thereby improving model robustness under incomplete data conditions. M3L [26] introduced a linear fusion mechanism combined with a multimodal teacher framework for masked modality learning, which enhances both semantic segmentation performance and model robustness under missing modality conditions. MMANet [27] improves representation learning by mining weak modal combinations and incorporating a regularization network to adaptively refine the model's response to incomplete modalities. Recent efforts have also explored advanced cross-modal reasoning strategies in broader multimodal domains. For instance, dynamic spatiotemporal graph reasoning has shown strong performance in video-based question answering tasks by capturing evolving temporal dependencies with self-supervised objectives [28]. Similarly, progressive perspective matching has been proposed for fine-grained cross-modal retrieval in remote sensing image-text scenarios [29]. These approaches underscore the increasing focus on higher order relational modeling and semantic alignment across modalities, providing valuable insight for improving fusion robustness and representation consistency in scenarios with incomplete data.

Despite these advancements, many existing methods still assume the availability of specific modalities during training. Furthermore, the ability of these models to maintain robustness under significant modality loss remains inadequately explored. This highlights the need for more flexible, robust, and scalable strategies in multimodal learning, particularly those that can effectively address the challenges posed by incomplete modalities in real-world applications.

III. Methodology

A. Problem Formulation

In a multimodal dataset with M modalities, the total number of possible modality combinations is given by 2^M-1. Fig. 3 illustrates seven different modality combinations derived from a trimodal ($M = 3$) dataset. The term "Combo" in Fig. 3 refers to modality combinations, with "Combo 1 - Combo 7" representing the seven possible combinations when $M = 3$. In this structure, two nodes are connected if their corresponding modality combinations share at least one common modality; otherwise, no connection is established between them.

In the task of multimodal semantic segmentation, a set of labeled training samples is considered, denoted as $D^{\text{train}} = \{D_l^0, D_l^1, \ldots, D_l^n\}$, where each sample is defined as $D_l^i = \{x_i^{\text{mod}_1}, x_i^{\text{mod}_2}, x_i^{\text{mod}_3}, y_i\}$ for $0 \le i \le n$. Here, $x_i^{\text{mod}_j}$ denotes the input data from modality mod_j, and y_i represents the corresponding ground-truth label. In the incomplete multimodal semantic segmentation setting, each sample must contain at least one available modality; that is, for each $x_i^{\text{mod}_j}$ with $j \in \{1, 2, 3\}$, not all modality inputs are zero or missing. Four distinct scenarios are defined based on the proportion of complete multimodal samples in the dataset: 100%, 75%, 50%, and 25%. The 100% setting indicates that all training and testing samples include data from all three modalities. In contrast, the 75% setting implies that only 75% of the samples contain all modalities, while the remaining 25% include at most two. The goal is to achieve robust semantic segmentation performance under these incomplete modality conditions by effectively leveraging the available modalities.

The principal challenge in incomplete multimodal fusion for remote sensing images lies in effectively learning and integrating multimodal features from incomplete modality combinations to ensure consistent and reliable performance across diverse scenarios. To address this problem, a multimodal heterogeneous hypergraph learning (MHHL) framework is proposed for incomplete semantic segmentation in remote sensing images. The training process is conducted on incomplete multimodal data $D_l^0, D_l^1, \ldots, D_l^n$. First, in each training batch, the incomplete multimodal data are subjected to feature extraction and fusion (Section III-B). Second, inputs with the same modality combinations are grouped as hypernodes to construct a multimodal heterogeneous hypergraph $\mathcal{G}$, with hyperedges established among them based on shared modality structures (Section III-C). The hypergraph structure enables feature updating and mapping for each hypernode through a transductive learning process (Section III-D), thereby facilitating mutual learning across different modality configurations. Finally, the aggregated features from various modality com-

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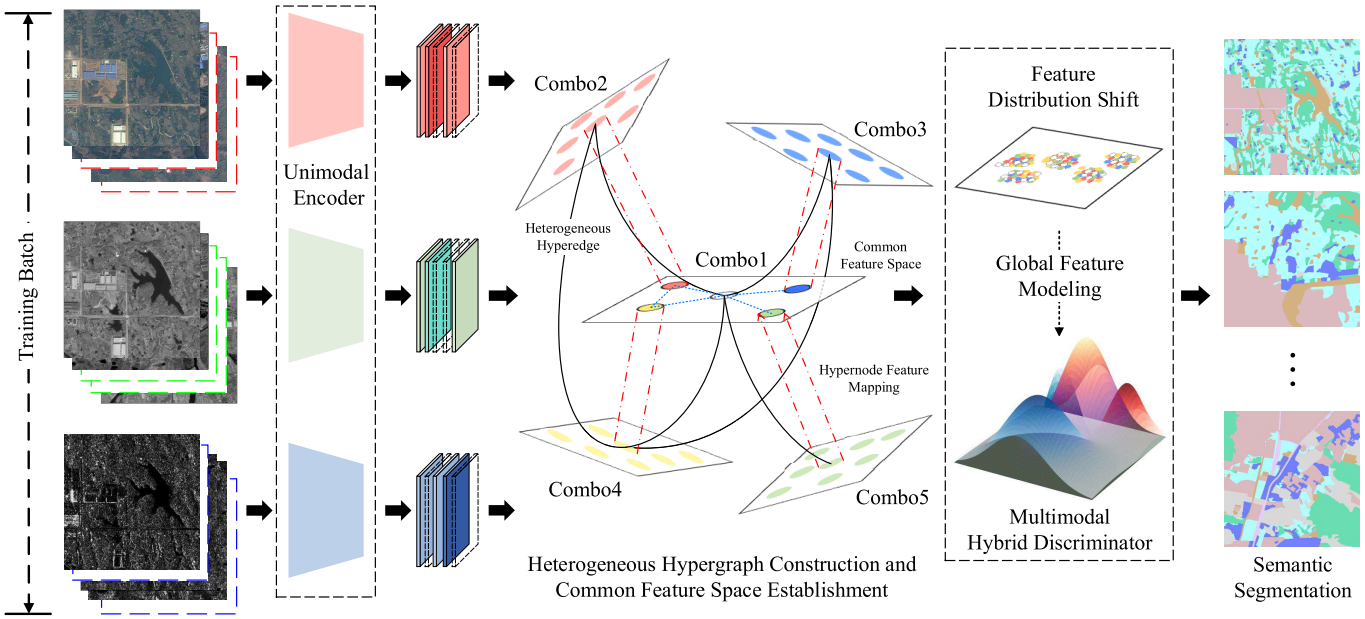


Fig. 2. Overview of the proposed MHHL framework for incomplete multimodal semantic segmentation.

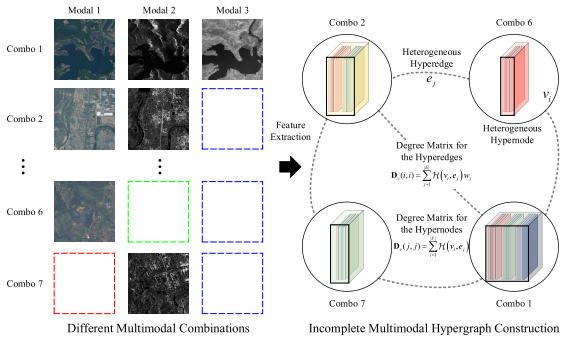


Fig. 3. Several different incomplete multimodal combinations and their corresponding relationships are represented within the multimodal heterogeneous hypergraph.

binations are passed through a multimodal mixed feature discriminator (Section III-E) for final prediction, completing the segmentation task under incomplete modality conditions. The workflow is summarized in Algorithm 1.

B. Hypernode Feature Extraction and Aggregation

The data belonging to different missing data modalities are located in different incomplete modalities with different feature spaces. If multimodal feature extraction and fusion are performed directly in the same feature space, it will lead to multimodal feature confusion. To address this challenge, a novel graph-theoretic structure is introduced, specifically designed to generalize the complexity of multimodal datasets with varying patterns of incomplete modalities. The graph structure, designated as the multimodal heterogeneous hypergraph (MHH), is designed to represent and correlate incomplete multimodal data efficiently through the utilization of complex hypergraph patterns.

First, the objective is to encode the incomplete multimodal data to extract both modality-specific features and cross-modal interaction information. The aggregated features obtained from this process are then regarded as hypernodes, laying the groundwork for the subsequent construction of the hypergraph. A hypernode-internal encoder is introduced to capture complementary information from highly interactive modalities, even in the presence of incomplete multimodal. The hypernode-internal encoder consists of two key components: 1) a unimodal feature extraction network that processes raw unimodal data and generates unimodal feature representations; and 2) an intranode feature interaction network that captures the intermodal interactions among these unimodal features, thus producing a hypernode-internal fusion feature representation.

Initially, a modality-specific feature extraction process is performed for each data source, resulting in the generation of distinct single-modality feature sets. These feature sets encapsulate the intrinsic characteristics inherent to each modality, thereby facilitating an initial comprehension of the underlying attributes of the ground objects as represented across the various modalities. The encoder architecture of the SegFormer [30] model is utilized to perform independent feature extraction for each modality. The SegFormer framework employs a distinctive Mobile Inverted Residual with Linear BottleNeck (MiT) module as its foundational network component. This module is strategically designed to systematically diminish the spatial resolution of the feature maps across sequential stages, concomitant with an incremental expansion of the channel dimensions. This orchestrated process facilitates the extraction of profound and intricate feature representations that encapsulate the nuanced characteristics of the ground objects within each distinct modality. Let I_m be the input image of the mth modality, and $F_m = \boldsymbol{E}_{\text{fea}}(I_m)$ is the feature of the mth modality extracted by the feature encoder, where $m = 1, 2, 3$.

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Algorithm 1 MHHL Under Incomplete Modalities

Input: Multimodal training set $\mathcal{D}_{train} = \{(I_i^m, y_i)\}$, testing set $\mathcal{D}_{test}$, maximum epochs E, hypergraph propagation steps T
Output: Trained model parameters Θ

```
Load Multimodal training set D_train = {(I_i^m, y_i)}
Initialize network parameters Θ
for e = 1 to E do
    for all batch B ⊂ D_train do
        /* Stage 1: Hypernode Feature Extraction*/
        for all modalities m in sample do
            Extract modality-specific features F_m = E_fea(I^m) via encoder
        end for
        Fuse features and construct hypernode features via (Eq. (2))
        /* Stage 2: Hypergraph Construction */
        Build heterogeneous hypergraph G = (V, E) based on modality availability
        Compute incidence matrix D and hyperedge weights W(Eq. (5)-(7))
        Compute adjacency matrix A (Eq. (8))
        /* Stage 3: Hypergraph Propagation*/
        for t = 0 to T − 1 do
            Update hypernode features H^(t+1) using spectral propagation (Eq. (9)-(11))
        end for
        /* Stage 4: Global Discriminative Modeling */
        Fit GMM using EM algorithm via Eq.(12)-(18)
        Perform class-wise GMM inference on features
        /* Stage 5: Loss Computation */
        Compute semantic segmentation loss
        Update model parameters Θ by minimizing total loss
    end for
end for
/* Testing Phase */
Evaluate trained model Θ on test set D_test and report accuracy
```

The fused feature representation F_{att} can be obtained by first applying an attention mechanism to weigh the contributions of each modality. This can be mathematically represented as

$$F_{\text{att}} = \sum_{m=1}^{3} \alpha_m \cdot F_m \tag{1}$$

where α_m represents the attention weight for the mth modality, which is learned during the network training process. These weights ensure that the most informative features from each modality are emphasized in the fusion process.

Next, a residual connection is introduced to add the original modality-specific features to the fused representation. The advantage of the residual mechanism becomes evident when a modality is missing. In such cases, the residual term helps maintain a strong baseline by retaining the contributions from the available modalities. This can be expressed as

$$\mathbf{H} = \mathbf{W}_h \left(\text{ReLU} \left(F_{att} + \sum_{m=1}^{3} \text{Conv1} \times 1 \left(F_m \right) \right) \right) + \mathbf{b}_h \tag{2}$$

where $\mathbf{H}$ is the matrix of aggregated hypernode features, and $\mathbf{W}_h$ and $\mathbf{b}_h$ are learnable parameters of the model, Conv1 $\times$ 1 is a 1 $\times$ 1 convolution used to learn the residual connections between different modalities adaptively. A hypernode encompasses multimodal features that interact with one another. The objective of the feature fusion process is to capture these interactions across different modalities, even under incomplete modality inputs, thereby extracting both modality-specific and cross-modality interaction information.

C. Multimodal Heterogeneous Hypergraph Construction

Upon obtaining the node features, we can commence the construction of the multimodal heterogeneous hypergraph model. Fig. 3 shows that these different combinations can be systematically grouped into distinct. Each node includes combinations that share a common missing modality, with the proviso that each combination is exclusively assigned to a single group on the basis of this criterion. Multimodal heterogeneous hypergraph (MHH) is denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ containing the following components and properties.

$\mathcal{V} = \{v_i\}_{i=1}^{N}$ is the set of nodes of a hypergraph where each hypernode is self-interacting. In contrast to hypergraph nodes with simple graphs, the nodes of a hypergraph contain a different number and dimension of features, and the features of each hypernode can interact implicitly or explicitly. In our problem, hypernodes refer to multimodal combinations.

$\mathcal{E} = \{e_j\}_{j=1}^{|\mathcal{E}|}$ is the set of edges. The set is used to represent higher-order connections between hypernodes more efficiently. In the field of graph theory, hypergraphs provide an extension to traditional graphs by allowing edges, termed hyperedges, to connect more than two vertices. Unlike standard graphs, where edges are limited to connecting pairs of vertices, hyperedges in a hypergraph can interconnect any number of vertices. In this article, two nodes are considered to be connected if they exhibit the same modality. Otherwise, they are not considered to be connected. This is illustrated in Fig. 3.

In multimodal learning tasks, each training batch may contain different incomplete modality combinations. Given the complex interplay between multimodal instances, i.e., hypernodes, each training batch provides a unique perspective on the relationships within the data. To capture the higher-order interactions among incomplete modality combinations, a strategy is adopted that first categorizes the data into distinct hypernode sets based on modality combinations. A set of hyperedges is then constructed to interconnect all hypernodes within each block. Let $\mathcal{V}_{\text{batch}}$ $(\forall v_i, v_j \in \mathcal{V}_{\text{batch}})$ denote the set of hypernodes involved in a block. The normalized distance between any two instances within the same combination is computed as follows:

$$d\left(v_i, v_j\right) = \left(1 - \frac{u\left(x_i\right) \cdot u\left(x_j\right)}{\|u\left(x_i\right)\|_2 \cdot \left\|u\left(x_j\right)\right\|_2}\right) \Big/ Z \tag{3}$$

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$$Z = \sum_{i,j \in \mathcal{V}_{\text{batch}}} \left(1 - \frac{u(x_i) \cdot u(x_j)}{\|u(x_i)\|_2 \cdot \|u(x_j)\|_2}\right) \tag{4}$$

where Z is the normalization factor and $u(\cdot)$ denotes node characteristics. Once distances are computed, each hyperedge is determined using the nearest neighbor approach centered on each node [31]. Through this method, combinations lacking a specific modality can be linked to combinations that do have one, thereby addressing the issue of incomplete data. In the figure on the right of Fig. 3, hyperedges of different colors indicate connections of different modal combinations.

Hypergraphs can be represented in various ways, one of which is through the use of an incidence matrix. This matrix is a 2-D array that delineates the relationships between the hypernodes and hyperedges of a hypergraph. The incidence matrix, denoted as $\mathcal{H}$, is defined as a $|V| \times |E|$ matrix, where $|V|$ represents the cardinality of the set of hypernodes, and $|E|$ represents the cardinality of the set of hyperedges. Each row in the matrix corresponds to a hypernode v_i, and each column corresponds to a hyperedge e_j. Entries within the incidence matrix $\mathcal{H}$ are binary, taking on values of either 0 or 1. The presence of a 1 at the entry $\mathcal{H}(v_i, e_j)$ signifies that the hypernode v_i is incident to the hyperedge e_j, indicating a connection through that hyperedge with other hypernodes in the graph. Conversely, a 0 at $\mathcal{H}(v_i, e_j)$ denotes the absence of such a connection. Furthermore, each hyperedge e_j is assigned a single-valued weight w_j. In the context of the article under discussion, it is stipulated that all edge weights w_j are equivalent to one, suggesting a uniform contribution of each hyperedge to the graph's structure

$$\mathbf{W}(j, j) = w_j = 1 \quad \forall j. \tag{5}$$

Following the definition of the incidence matrix $\mathcal{H}$, we can derive the degree matrices $\boldsymbol{D}_v$ (for hypernodes), $\boldsymbol{D}_e$ (for hypernodes), and the adjacency matrix $\mathbf{A}$. The degree matrix $\boldsymbol{D}_v$ is a diagonal matrix representing the degree of each hypernode, where the degree is defined as the sum of the weights of hyperedges incident to the hypernode. Formally, the degree matrix $\boldsymbol{D}_v$ is given by

$$\boldsymbol{D}_v(i, i) = \sum_{j=1}^{|E|} \mathcal{H}(v_i, e_j) w_j. \tag{6}$$

Similarly, the degree matrix for the hyperedges, denoted as $\boldsymbol{D}_e$, is a diagonal matrix where each diagonal entry $\boldsymbol{D}_e(j, j)$ represents the sum of the weights of hypernodes incident to the hyperedge e_j. It is defined as

$$\boldsymbol{D}_e(j, j) = \sum_{i=1}^{|V|} \mathcal{H}(v_i, e_j). \tag{7}$$

The adjacency matrix $\mathbf{A}$ of the hypergraph encodes the pairwise relationships between hypernodes based on their shared hyperedges. It is an $|V| \times |V|$ matrix, where each entry $\mathbf{A}(v_i, v_j)$ represents the sum of the products of the weights of hyperedges that connect hypernodes v_i and v_j. The adjacency matrix $\mathbf{A}$ can be expressed as

$$\mathbf{A} = \mathcal{H}\mathbf{W}\mathcal{H}^T - \boldsymbol{D}_v \tag{8}$$

where $\mathbf{W}$ is the diagonal matrix of hyperedge weights and $\boldsymbol{D}_v$ is the degree matrix for the hypernodes. The subtraction of $\boldsymbol{D}_v$ ensures that self-loops (where $i = j$) are properly handled, typically by setting them to zero.

Using the incidence matrix $\mathcal{H}$, the adjacency matrix $\mathbf{A}$, degree matrices $\boldsymbol{D}_v$ and $\boldsymbol{D}_e$, and the weight matrix $\mathbf{W}$, we can effectively represent the hypergraph's structure and relationships. This representation allows for various operations on the hypergraph, such as node feature mapping, feature propagation, and update mechanisms. By leveraging these matrices, the hypergraph's structure can be utilized to facilitate complex interactions among hypernodes, allowing for the effective extraction of both modality-specific and cross-modal features in scenarios involving incomplete multimodal data.

D. Hypernode Feature Mapping and Hypergraph Learning

After constructing the hypergraph $\mathcal{G}$, our primary objective is to utilize the graph structure to enable mutual learning across various incomplete modality combinations. We exploit the graph structure to conduct feature mapping and updating processes, which facilitate interactive learning among the nodes. This approach is critical for capturing both modality-specific characteristics and cross-modal interactions, especially in scenarios where certain modalities are missing. The process unfolds in two primary stages:

1) Hypernode Feature Mapping: The first stage involves mapping the aggregated features of each hypernode into a lower dimensional space. This dimensionality reduction is designed to distill the most pertinent information while preserving the inherent structural relationships within the hypergraph.

Let $\mathbf{H} \in \mathbb{R}^{n \times d}$ represent the matrix of aggregated hypernode features, where n is the number of hypernodes and d is the dimensionality of the feature space. The goal is to project $\mathbf{H}$ into a lower-dimensional space $\mathbb{R}^{n \times d'}$ (with $d' < d$) using a series of graph convolutional layers. The transformation for each hypernode i is given by

$$\mathbf{h}_i' = \text{ReLU}\left(\sum_{j \in \mathcal{N}(i)} \mathbf{W}_1 \mathbf{h}_j + \mathbf{b}_1\right) \tag{9}$$

where $\mathbf{h}_i'$ is the lower dimensional feature vector for hypernode i, $\mathbf{h}_j$ represents the feature vector of a neighboring node j in the hypergraph $\mathcal{G}$, $\mathbf{W}_1 \in \mathbb{R}^{d \times d'}$ is a trainable weight matrix for the graph convolutional layer, $\mathbf{b}_1 \in \mathbb{R}^{d'}$ is a bias term, $\mathcal{N}(i)$ denotes the set of neighbors of node i within the hypergraph. This mapping process ensures that the lower dimensional feature representations $\mathbf{H}'$ maintain the essential structural properties and are capable of capturing both intra- and intermodal interactions.

2) Feature Update via Graph Convolution: Following the dimensionality reduction, the next stage focuses on updating the hypernode features using a Graph Convolutional Network (GCN). The purpose of this update is to refine the feature representations by leveraging the connectivity of the hypergraph, enabling the propagation of both local and global contextual

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information. The update step for each hypernode i is expressed as

$$\mathbf{h}_i^{(t+1)} = \mathrm{ReLU}\left(\sum_{j\in\mathcal{N}(i)} \mathbf{W}_2^{(t)}\mathbf{h}_j^{(t)} + \mathbf{b}_2^{(t)}\right) \tag{10}$$

where $\mathbf{h}_i^{(t+1)}$ is the updated feature vector for hypernode i at iteration $t+1$, $\mathbf{W}_2^{(t)} \in \mathbb{R}^{d'\times d'}$ is a trainable weight matrix for the GCN at iteration t, $\mathbf{b}_2^{(t)} \in \mathbb{R}^{d'}$ is a bias term for the same iteration.

The iterative application of this graph convolution allows for the progressive refinement of hypernode features, ensuring that each hypernode incorporates information from its neighbors and thus enhances the overall robustness and discriminative power of the feature representations. It can be further expanded using the following general formulation:

$$\mathbf{H}^{(t+1)} = \mathrm{ReLU}\left(\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\tilde{\mathbf{A}}\tilde{\boldsymbol{D}}^{-\frac{1}{2}}\mathbf{H}^{(t)}\mathbf{W}_2^{(t)}\right) \tag{11}$$

where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix of the hypergraph with added self-loops, $\tilde{\boldsymbol{D}}$ is the degree matrix corresponding to $\tilde{\mathbf{A}}$, $\mathbf{H}^{(t)}$ represents the feature matrix at iteration t. The iterative process of feature mapping and updating allows the hypernode features to be progressively enriched, leading to final representations. The ability to refine features iteratively ensures that the model is resilient to the challenges posed by missing modalities and can effectively harness the complementary strengths of available data. The final feature representation $\mathbf{H}^{(T)}$, after T iterations, encapsulates both modality-specific details and cross-modal interactions, providing a robust foundation for high-level tasks in scenarios characterized by incomplete multimodal data.

In summary, this iterative approach enables the hypergraph to capture and refine critical multimodal features through a combination of dimensionality reduction and graph convolution, ensuring the generation of robust and discriminative feature representations. For theoretical justification of the proposed hypergraph-based feature propagation and its robustness under incomplete multimodal configurations, please refer to Appendix A.

E. Multimodal Global Hybrid Modeling

In incomplete multimodal semantic segmentation, the variability in available modality combinations can lead to feature shift issues. Each modality offers unique information about the ground objects, and the absence of certain modalities in some data instances may result in inconsistencies in the aggregated feature distribution. To address this problem, the proposed approach focuses on global multimodal feature modeling by introducing a multimodal mixed feature discriminator. Bayesian inference is employed to facilitate label prediction. Specifically, the generative classifier models the joint probability distribution $p(x, c)$ and estimates the class-conditional distribution $p(x|c)$ in conjunction with the class prior $p(c)$. Subsequently, the posterior probability $p(c|x)$ is derived in accordance with Bayes' theorem. We integrate the generative expectation-maximization (EM) optimization of Gaussian mixture models with the discriminative learning of dense feature extractors. This amalgamation gives rise to a robust, principled framework for segmentation grounded in dense generative classification. Specifically, it employs a weighted mixture of M multivariate Gaussians to model the distribution of pixel data for each class c in the D-dimensional embedding space

$$\begin{aligned} p\left(\boldsymbol{x} \mid c; \boldsymbol{\phi}_c\right) &= \sum_{m=1}^{M} p\left(m \mid c; \boldsymbol{\pi}_c\right) p\left(\boldsymbol{x} \mid c, m; \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c\right) \\ &= \sum_{m=1}^{M} \pi_{cm} \mathcal{N}\left(\boldsymbol{x}; \boldsymbol{\mu}_{cm}, \boldsymbol{\Sigma}_{cm}\right) \end{aligned} \tag{12}$$

where $m \mid c \sim \mathrm{Multinomial}(\boldsymbol{\pi}_c)$ represents the prior probability, with the constraint $\sum_m \pi_{cm} = 1$; $\boldsymbol{\mu}_{cm} \in \mathbb{R}^d$ and $\sum_{cm} \in \mathbb{R}^{d\times d}$ are the mean vector and covariance matrix of component mm for class c, respectively; $\boldsymbol{\phi}_c = \{\boldsymbol{\pi}_c, \boldsymbol{\mu}_c, \boldsymbol{\Sigma}_c\}$. The mixture nature endows the discriminator with the capability to approximate data densities accurately. Each Gaussian component possesses an independent covariance structure, allowing for flexible and localized measurement of importance along different feature dimensions. To identify the optimal parameters for the classifier, that is, to satisfy the constraint $\{\boldsymbol{\phi}_c^*\}_{c=1}^C$ the standard approach is the EM algorithm [32], which maximizes the log-likelihood of the feature-label pairs $\{(\boldsymbol{x}_n, c_n)\}_{n=1}^N$ over the training sample set D with N observations

$$\begin{aligned} \boldsymbol{\phi}_c^* &= \arg\max_{\boldsymbol{\phi}_c} \sum_{\boldsymbol{x}_n: c_n = c} \log p\left(\boldsymbol{x}_n \mid c; \boldsymbol{\phi}_c\right) \\ &= \arg\max_{\boldsymbol{\phi}_c} \sum_{\boldsymbol{x}_n: c_n = c} \log \sum_{m=1}^{M} p\left(\boldsymbol{x}_n, m \mid c; \boldsymbol{\phi}_c\right). \end{aligned} \tag{13}$$

The EM algorithm initiates with some initial guesses for the maximum likelihood parameters $\lambda_c^{(0)}$ and then iteratively constructs successive estimates $\lambda_c^{(t)}$ for $t = 1, 2, \ldots$, by repeatedly optimizing the function

$$E\text{-Step: } q_c^{(t)} = \arg\max_{q_c} F\left(q_c, \boldsymbol{\phi}_c^{(t-1)}\right) \tag{14}$$

$$M\text{-Step: } \boldsymbol{\phi}_c^{(t)} = \arg\max_{\boldsymbol{\phi}_c} F\left(q_c^{(t)}, \boldsymbol{\phi}_c\right) \tag{15}$$

$q_c[m] = p(m \mid \boldsymbol{x}, c; \boldsymbol{\phi}_c)$ provides the likelihood of the data $\boldsymbol{x}$ being assigned to component m. The function F is defined as

$$F\left(q_c, \boldsymbol{\phi}_c\right) = \mathbb{E}_{q_c}\left[\log p\left(\boldsymbol{x}, m \mid c; \boldsymbol{\phi}_c\right)\right] + H\left(q_c\right) \tag{16}$$

where $\mathbb{E}_{q_c}[\cdot]$ denotes the expectation with respect to the distribution over the M components as prescribed by q_c, and $H(q_c) = -\mathbb{E}_{q_c}[\log q_c[m]]$ delineates the entropy of q_c. In the E-step, we recompute the posterior $q_c^{(t)}$ over the M components given the old parameters. In the M-step, leveraging the soft cluster assignments $q_c^{(t)}$, the parameters are updated to $\phi_c^{(t)}$ to maximize the function F.

The classification process proceeds by utilizing the trained parameters $\boldsymbol{\phi}_c^*$ obtained from the EM algorithm to predict the class labels for new data instances. For a given input $\boldsymbol{x}$, the posterior probability of class c given $\boldsymbol{x}$ is computed using Bayes' theorem

$$p\left(c \mid \boldsymbol{x}\right) = \frac{p\left(\boldsymbol{x} \mid c; \boldsymbol{\phi} c^*\right) p\left(c\right)}{\sum c' p\left(\boldsymbol{x} \mid c'; \boldsymbol{\phi}_{c'}^*\right) p\left(c'\right)} \tag{17}$$

where $p(\boldsymbol{x} \mid c; \boldsymbol{\phi}_c^*)$ is the class-conditional probability density function, modeled as a Gaussian mixture, and $p(c)$ is the prior

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TABLE I
SAMPLE NUMBER OF TRAINING DATA AND TEST DATA UNDER DIFFERENT DATA RATIOS

Dataset	WHU				Hunan			
Type / Ratios	Train		Test		Train		Test	
	Complete	Incomplete	Complete	Incomplete	Complete	Incomplete	Complete	Incomplete
100%	5632	0	1408	0	400	0	100	0
75%	4224	1408	1056	352	300	100	75	25
50%	2816	2816	704	704	200	200	50	50
25%	1408	4224	352	1056	100	300	25	75

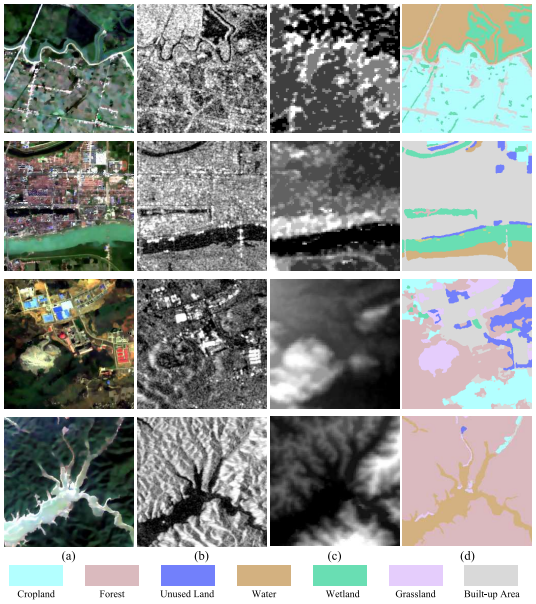


Fig. 4. Partial image visualization of the Hunan dataset is presented here. The images are presented in the following order. (a) Multispectral images. (b) SAR images. (c) SRTM images. (d) Ground-truth images. Each row represents a set of multimodal images and labels.

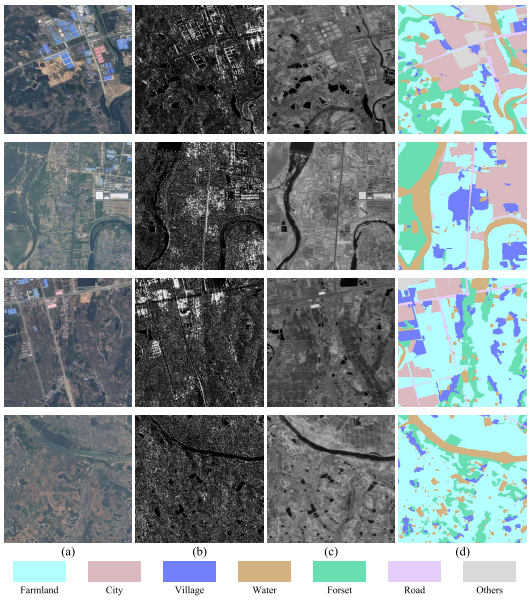


Fig. 5. Partial visualization of the WHU dataset is presented here. (a)–(d) RGB optical images, SAR images, infrared images, and ground-truth images, respectively, in the order presented. Each row represents a set of multimodal images and labels.

probability of class c. The predicted class label $\hat{c}$ for the input $\boldsymbol{x}$ is then determined by maximizing the posterior probability

$$\hat{c} = \arg\max_{c} p\left(c \mid \boldsymbol{x}\right). \tag{18}$$

During training, the classifier is refined by iteratively applying the EM algorithm to adjust the parameters $\boldsymbol{\phi}_c$ for each class. This iterative refinement enhances the model's ability to capture the underlying distribution of the data and improves classification accuracy. The model learns to distinguish between classes based on the aggregated features derived from the multimodal inputs, effectively handling the feature shift issue caused by incomplete modality combinations.

IV. EXPERIMENT

In this section, extensive experiments are conducted on two widely used image classification datasets: the Hunan dataset [33] and the WHU dataset [12].

A. Datasets

1) Hunan Dataset: The *Hunan dataset* [33] is a tri-modal dataset created and labeled by the Experimental Team of Skyearth in the School of Remote Sensing and Information Engineering, Wuhan University, China. The dataset consists of multispectral (MSI), SAR, raw elevation layers, and slope layers derived from DEM data collected by Sentinel-1, Sentinel-2, and the Shuttle Radar Topography Mission (SRTM), respectively. Sentinel-2 collects MSI data with 13 channels, Sentinel-1 collects SAR data in both VV and VH polarizations, and the SRTM data have both elevation and slope sources. The dataset delineates seven land cover types, including farmland, forest, grassland, wetland, watershed, unutilized land, and built-up area, as shown in Fig. 4. The dataset contains 500 images of 256×256 size and includes corresponding reference semantic labels at 10 m spatial resolution. The number of training data and test data in each case is shown in Table I.

2) Whu Dataset: The *WHU dataset* [12] consists of 100 optical images with a resolution of 5556×3704 pixels, along with SAR images captured over the same region. This dataset spans approximately 50000 km^2 in Hubei Province, China (30°N–33°N, 108°E–117°E). For analysis, the large images are divided into 7040 nonoverlapping 512×512 patches. The dataset covers seven distinct land cover categories, as

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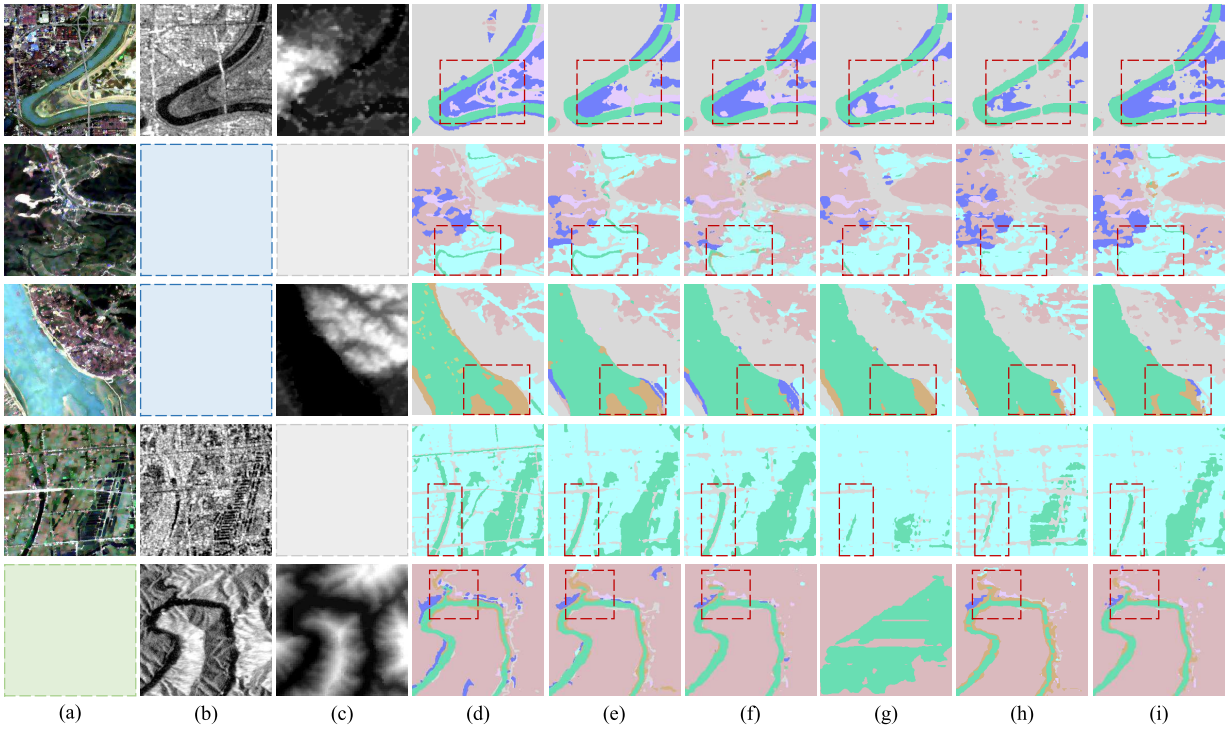


Fig. 6. Results of semantic segmentation obtained using different approaches for the Hunan dataset. (a) MSI image. (b) SAR image. (c) SRTM image. (d) Ground-truth image. (e) Ours. (f) CMNeXt. (g) M3L. (h) MMANet. (i) MANet.

illustrated in Fig. 5, and the distribution of training and test data for each case is provided in Table I.

B. Experimental Setup

The MHHL model is implemented using the PyTorch framework. The backbone is based on the Mix Transformer (MiT) encoder pretrained on ImageNet, with an embedding dimension of 512, as introduced in SegFormer [30]. We use the AdamW optimizer with a weight decay of 0.01, an initial learning rate of 6×10^{-5}, and a polynomial learning rate schedule. The model is trained for 200 epochs on a single 4090 GPU, with a batch size of 4 for the WHU dataset and 8 for the Hunan dataset.

To evaluate the effectiveness of the MHHL model in incomplete multimodal semantic segmentation, we conduct both visual and quantitative analyses. Four distinct incomplete multimodal scenarios are set up, as shown in Table I. In the 100% scenario, all modalities are present, while in the 75% scenario, 75% of the data have all three modalities, and the rest contain at most two modalities. The performance of the model is assessed under both complete and incomplete data conditions. The MHHL model is compared with several representative multimodal fusion methods, including CMX [19], CMNeXt [34], Dformer [20], FTransUNet [35], MANet [36], MMSFormer [13] as well as M3L [26] and MMANet [27], which are specifically designed. These methods are combined with the same backbone to benchmark their performance against the proposed approach.

Several widely used metrics were employed for the quantitative evaluation, including class accuracy (CA), overall accuracy (OA), mean intersection over union (mIoU), and frequency-weighted IoU (Freq_IoU). The intersection over union (IoU) measures the overlap between the predicted and true segmentation maps, indicating the accuracy of the model's predictions. The mIoU calculates the average IoU across all categories, offering an overall performance evaluation of the model. Freq_IoU considers the occurrence frequency of each category across all images, reducing the influence of common categories on the performance, thus providing a more balanced evaluation, especially for less frequent categories. OA represents the proportion of correctly classified pixels relative to the total number of pixels, though it does not differentiate between categories, focusing solely on the overall correctness of classification. These metrics offer a comprehensive assessment of the model's segmentation performance, with higher values indicating better accuracy in remote sensing image classification tasks.

C. Experimental Results on Hunan Dataset

The Hunan dataset consists of 400 training images and 100 test images, representing seven land cover classes: farmland, forest, grassland, wetland, watershed, unutilized land, and built-up areas, as shown in Fig. 4. Table II presents a performance comparison of various algorithms on this dataset, with the last row highlighting the superior results of our proposed MHHL network, which outperforms others in OA, mIoU, and Freq_IoU. Additionally, Fig. 6 showcases visualized segmentation results from our method and other comparative models, providing a more detailed examination of the results.

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TABLE II
COMPARISON OF THE RESULTS OF DIFFERENT MULTIMODAL FUSION METHODS ON THE HUNAN DATASET

Method	100%			75%			50%			25%		
	mIoU(%)	Freq IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)
CMX [19]	46.81	64.28	76.44	37.70	53.86	69.19	35.56	48.17	63.84	29.16	41.34	57.64
CMNeXt [34]	46.81	65.20	78.79	45.47	60.75	75.43	43.17	57.10	71.72	34.36	48.55	65.03
Dformer [20]	47.03	65.93	78.73	41.49	55.83	70.89	36.83	50.92	66.70	35.16	46.52	62.01
FTransUNet [35]	44.74	58.32	73.18	36.25	55.90	71.46	39.73	51.98	67.59	33.45	47.90	64.21
MANet [36]	45.78	65.34	79.10	41.97	59.54	74.67	42.88	59.03	73.36	39.52	52.73	68.79
MMSFormer [13]	48.57	65.97	79.21	43.21	62.14	76.71	43.28	57.40	72.07	37.55	52.15	67.52
M3L [26]	-	-	-	37.08	51.04	66.62	34.95	46.13	61.82	27.79	41.15	57.40
MMANet [27]	-	-	-	44.23	59.64	73.89	38.94	53.78	69.32	35.25	50.25	67.45
Ours	**52.69**	**69.28**	**81.23**	**48.70**	**64.05**	**77.49**	**44.58**	**61.25**	**75.75**	**43.63**	**56.85**	**71.41**

Table II presents a performance comparison of various methods on the Hunan dataset, evaluated using key metrics including mIoU, Freq_IoU, and OA. These metrics were calculated under 100%, 75%, 50%, and 25% incomplete modality settings to assess the model's performance under varying levels of modality availability. As shown in Table II, our proposed method outperforms others in all evaluation metrics. For mIoU, our method achieves 52.69% with 50% modal missing, 69.28% with 75% modal missing, and 81.23% with complete data, demonstrating its ability to accurately identify and segment categories. For Freq_IoU, we achieve 48.70% under 50% modal missing, 64.05% under 75% modal missing, and 77.49% with full data, indicating a balanced handling of categories with varying frequencies. The OA metrics show similar trends, with our method reaching 44.58% under 50% modal missing, 61.25% under 75% modal missing, and 75.75% with complete data. By comparison, other methods such as CMNeXt, Dformer, M3L, and MMANet exhibit relatively lower performance in terms of mIoU, Freq_IoU, and OA. It highlights the advantages of our method in multimodal data fusion and semantic segmentation. While CMNeXt performs well with complete data, its performance significantly drops with missing modalities in both training and test data. Specifically, our method shows a decrease of 9.82% in OA and 9.06% in mIoU under 25% modal missing, whereas CMNeXt's performance drops by 13.76% in OA and 12.45% in mIoU. Other methods, such as CMX and Dformer, show a sharp decline in all metrics, struggling with modal absence. Even in the absence of certain modalities, the MANet retains the ability to extract semantically meaningful and stable features from the available inputs. The pretrained knowledge of SAM, combined with the modality-adaptive nature of MMAdapter, ensures that MANet maintains high segmentation performance despite partial input data.

Fig. 6 presents the semantic segmentation results of different methods on the Hunan dataset under a 50% incomplete multimodal setting. Specifically, (a), (b), and (c) show the input images from the MSI, SAR, and SRTM elevation modalities, respectively, while (d) shows the corresponding ground truth. The segmentation results produced by our method, CMNeXt, M3L, MMANet, and MANet are displayed in subfigures (e) through (i), respectively. Regions without corresponding input images indicate missing modalities during the testing phase. To better illustrate the effectiveness of the proposed method, red bounding boxes are used to highlight key regions where our approach demonstrates noticeable improvements over other methods. These areas emphasize enhanced boundary delineation and more accurate object recognition, particularly under modality-missing conditions. The visual results clearly reflect the robustness and superiority of our method in handling incomplete multimodal data.

D. Experimental Results on Whu Dataset

The WHU dataset contains 5632 training images and 1408 test images, covering various land cover classes such as farmland, urban areas, villages, water bodies, forests, roads, and other features, totaling seven categories. Example images are shown in Fig. 5. The experimental setup is described above. Table III compares the performance of different algorithms on the WHU dataset, with the last row highlighting the results of our proposed MHHL network. Our method achieves the highest OA, mIoU, and frequency-weighted IoU, outperforming other approaches. Fig. 7 visualizes the experimental results, providing further insight into the comparative performance.

Table III compares the performance of different methods in terms of key metrics, including mIoU, Freq_IoU, and OA. These metrics were evaluated at 100%, 75%, 50%, and 25% modal absence to assess performance under varying levels of missing modalities. As shown in Table III, our proposed method excels across all metrics. For mIoU, we achieved 48.99% with 50% modal missing, 69.89% with 75% modal missing, and 60.51% with full data, indicating accurate segmentation across categories. For Freq_IoU, our method performed well with 64.10% at 50% incomplete modal, 69.89% at 75% incomplete modal, and 74.28% with complete data, highlighting the model's effectiveness in balancing categories of varying frequencies. OA results also show strong performance, with 77.61% at 50% incomplete modal, 81.88% at 75% incomplete modal, and 84.99% under full data.

In contrast, other methods, such as CMNeXt, Dformer, M3L, and MMANet, fall short in mIoU, Freq_IoU, and OA metrics, despite achieving good results overall. While CMNeXt achieves strong results with full modalities, its performance degrades significantly under modal missing, showing a 23.13% drop in OA and 28.14% in mIoU with 25% incomplete data, compared to our method's smaller decline of 14.55% and 19.58%, respectively. CMX and Dformer also

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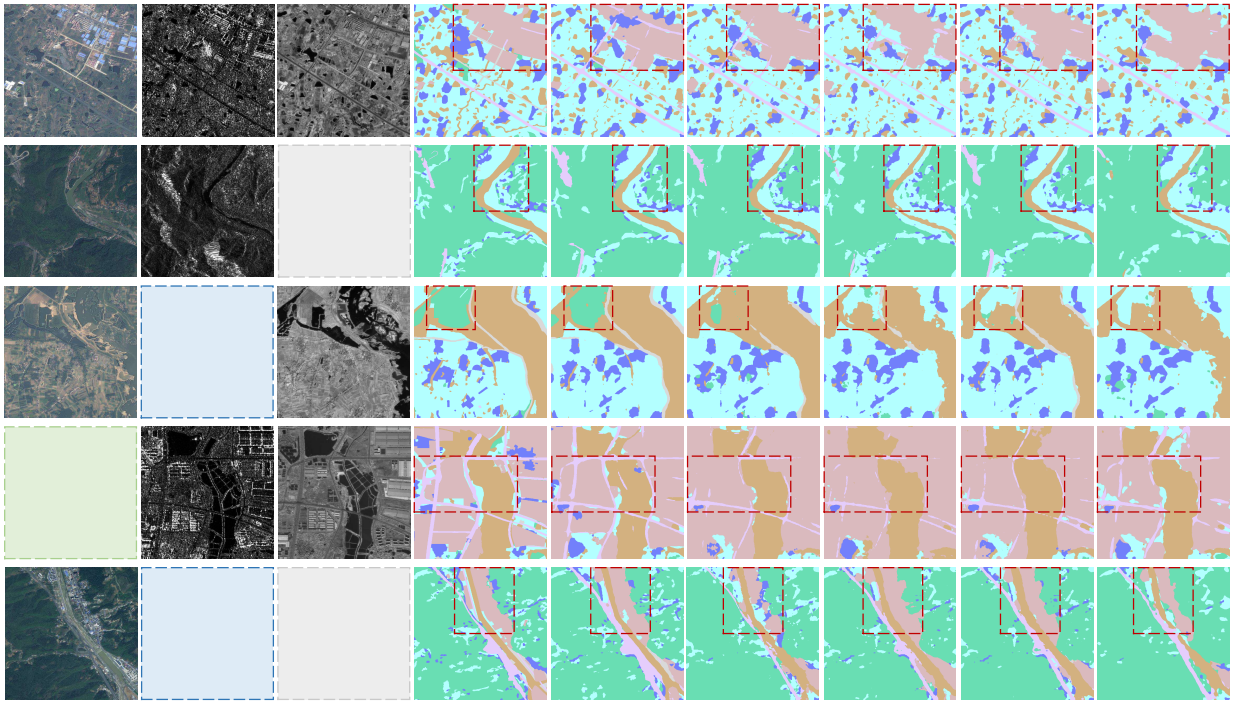


Fig. 7. Results of semantic segmentation obtained using different approaches for the WHU dataset. (a) MSI image. (b) SAR image. (c) SRTM image. (d) Ground-truth image. (e) Ours. (f) CMNeXt. (g) M3L. (h) MMANet. (i) MANet.

TABLE III
COMPARISON OF THE RESULTS OF DIFFERENT MULTIMODAL FUSION METHODS ON THE WHU DATASET

Method	100%			75%			50%			25%		
	mIoU(%)	Freq_IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)	mIoU(%)	Freq_IoU(%)	OA(%)
CMX [19]	56.73	71.84	83.25	49.39	62.98	76.73	40.18	55.01	69.72	27.58	41.20	57.77
CMNeXt [34]	58.24	73.15	84.26	55.70	65.07	78.94	45.02	57.21	72.05	30.10	43.37	61.13
Dformer [20]	55.51	71.40	82.99	47.09	61.65	75.37	38.34	51.82	67.11	34.08	44.11	58.51
FTransUNet [35]	49.41	67.44	80.17	42.23	58.90	74.23	39.21	55.07	70.89	31.41	44.06	61.89
MANet [36]	58.71	72.92	84.03	49.61	66.91	79.89	42.23	58.90	74.23	36.11	50.39	67.34
MMSFormer [13]	57.55	72.29	83.62	46.23	66.23	79.11	44.47	59.74	74.29	34.45	48.84	65.87
M3L [26]	-	-	-	46.47	59.51	73.41	36.81	49.69	64.99	26.64	40.57	56.55
MMANet [27]	-	-	-	52.84	63.68	77.72	45.47	59.97	73.83	33.56	43.36	58.99
Ours	**60.51**	**74.28**	**84.99**	**53.92**	**69.89**	**81.88**	**48.99**	**64.10**	**77.61**	**40.93**	**56.05**	**70.44**

suffer marked reductions across all metrics, highlighting their vulnerability to incomplete modalities. MMANet benefits from its MAD and MAR modules, showing improved robustness as modal availability decreases.

Fig. 7 illustrates segmentation results under 50% incomplete modality. Fig. 7(a)–(d) show the MSI, SAR, SRTM, and ground truth, respectively; (e)–(i) present results from our method, CMNeXt, M3L, MMANet, and MANet. Missing images indicate absent modalities during testing. Compared to others, our method preserves structural integrity and achieves higher accuracy despite incomplete inputs.

E. Experimental Results of Multimodal Analysis

To comprehensively evaluate the performance of our proposed method under different combinations of modal data, we designed a series of simulation validation scenarios covering different cases from partially modal missing to full modal data. Specifically, we considered the scenarios with 50% modal missing, 50% full modal training data, MSI data only, SAR data only, and SRTM data only. Table IV and Fig. 8 provide the experimental results for these scenarios.

TABLE IV
EXPERIMENTAL RESULTS OF OUR METHOD WITH DIFFERENT COMBINATIONS OF MODAL DATA

Data	mIoU(%)	Freq_IoU(%)	OA(%)
50% Incomplete Multimodal	44.58	61.25	75.75
50% Multimodal	41.84	57.01	71.74
Only MSI	45.58	59.06	73.32
Only SAR	32.69	53.73	70.71
Only SRTM	21.46	43.23	61.46

Table IV shows the results for the mIoU, Freq_IoU, and OA for different combinations of modal data. Our method achieves 44.58% mIoU, 61.25% Freq_IoU, and 75.7% OA with 50% incomplete multimodal data, demonstrating strong

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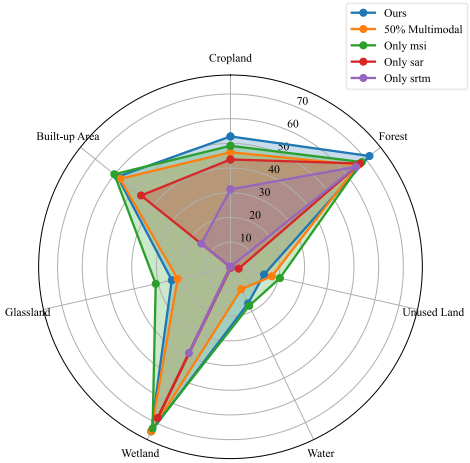


Fig. 8. Radar plots of the experimental results for each category under different combinations of modal data.

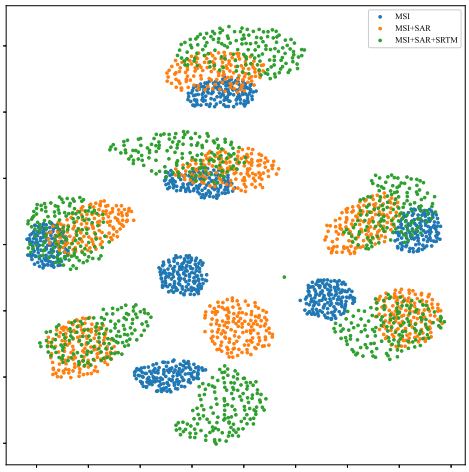


Fig. 9. Distribution of features for different modal combinations.

TABLE V
ABLATION EXPERIMENT RESULTS OF HYPERGRAPHIC LEARNING

Nodes	Graph	mIoU(%)	Freq IoU(%)	OA(%)
✓		38.04	52.98	67.65
	✓	41.89	57.46	71.39
✓	✓	43.18	58.13	73.62

segmentation performance under modality deficiency. In contrast, using only 50% of the full-modal training data results in lower performance (41.84% mIoU, 57.01% Freq_IoU, 71.74% OA), highlighting the effectiveness of our hypergraph-based framework in leveraging incomplete information. When using only MSI, SAR, or SRTM individually, performance drops significantly—MSI: 45.58% mIoU, SAR: 32.69%, SRTM: 21.46%—indicating that single modalities lack sufficient discriminative information for accurate segmentation.

Fig. 8 presents per-CA under different modality settings. The radar plot shows that Cropland, Built-up Area, and Forest achieve higher accuracy under 50% multimodal input, while Wetland and Water remain challenging, likely due to weaker modality-specific representations.

F. Ablation Experiments for Hypergraphic Learning

To evaluate the effectiveness of the proposed hypergraph learning approach under 50% incomplete multimodal, a series of ablation experiments were conducted, as shown in Table V. The results demonstrate that our multimodal heterogeneous hypergraph and graph-based direct push learning framework substantially improves semantic segmentation performance.

Table V summarizes the ablation results for different hypergraph node configurations. Without a hypergraph structure, the model achieves 38.04% mIoU, 52.98% Freq_IoU, and 67.65% OA. Introducing supernode mapping improves performance to 41.89% mIoU, 57.46% Freq_IoU, and 71.39% OA. With the complete hypergraph, the results further improve to 43.18% mIoU, 58.13% Freq_IoU, and 73.62% OA. These findings validate the effectiveness of the proposed multimodal heterogeneous hypergraph in capturing complex cross-modal relationships. By leveraging heterogeneous hypernodes and hyperedges, the model captures modality-specific interactions and complements missing modalities with contextual cues. Additionally, the graph neural network-based direct push learning framework facilitates cross-modal knowledge transfer, enabling robust feature learning under incomplete modality conditions.

G. Ablation Experiments for Multimodal Decoder

The present study conducts a comprehensive evaluation of the proposed decoder and verifies its effectiveness in multimodal semantic segmentation through a series of comparative experiments. As depicted in Fig. 9, the feature distributions extracted from various modality combinations—after being processed by the hypergraph network and prior to entering the decoder—exhibit distinct patterns. Specifically, the distributions corresponding to the "MSI" and "MSI+SAR" combinations reveal notable dispersion and inconsistency, whereas the "MSI+SAR+SRTM" combination yields a more compact and uniform distribution. These observations indicate the presence of modality-induced feature bias, wherein discrepancies among feature representations originating from different modality fusions may impair the decoder's performance. The representational imbalance caused by such heterogeneity could undermine the model's ability to generalize across varying input configurations. To address this issue, we propose a multimodal hybrid discriminative decoder.

Table VI presents the comparative performance of various decoders under identical network training conditions, accompanied by the mIoU, Freq_IoU, and OA. The proposed method demonstrates high performance across all categories, with particularly impressive results observed for the "Cropland" and "Forest" categories, achieving 52.80% and 71.97% mIoU. Furthermore, the overall performance is also excellent, with 44.58% mIoU, 61.25% Freq_IoU, and 75.75% OA. Conversely, conventional decoders, such as the multilayer perceptron (MLP), fully convolutional network (FCN), and U-Net Improved (Upernet), do not perform as well as our

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TABLE VI
COMPARATIVE EXPERIMENTS WITH DIFFERENT MULTIMODAL DECODERS

Class \ Decoder	**Ours**	MLP	FCN	Upernet
Cropland	52.80	42.65	43.46	**53.28**
Forest	**71.97**	59.03	65.00	68.51
Unused Land	13.95	17.25	8.41	**26.34**
Water	16.44	1.13	8.61	**16.56**
Wetland	**73.98**	64.53	71.31	61.15
Grassland	24.38	23.73	20.81	**28.61**
Built-up Area	**58.50**	49.40	46.21	47.83
mIoU(%)	**44.58**	36.83	37.70	43.18
Freq_IoU(%)	**61.25**	50.92	53.86	58.13
OA(%)	**75.75**	66.70	69.19	73.62

method in certain categories. For instance, the MLP achieved a mIoU of only 1.13% in the "Water" category, whereas our method achieved 16.44%. The FCN achieved a mIoU of 8.42% in the "UnusedLand" category, while our method achieved 13.95%. However, this result is still slightly inferior to our method.

Table VI demonstrates that the proposed multimodal hybrid discriminative model effectively alleviates feature distribution bias across different modality combinations. It substantially enhances the performance of multimodal semantic segmentation. Moreover, the multimodal Gaussian hybrid feature discriminant strengthens the model's capacity for feature fusion and discrimination. As a result, the model achieves high segmentation accuracy and robust performance across various modality configurations.

H. Discussion: Failure Cases and Limitations

While the proposed hypergraph-based feature modeling framework exhibits strong performance under incomplete multimodal conditions, several limitations persist.

1) Modality Sparsity and Imbalance: The approach relies on modality overlap to construct hyperedges and propagate information. In cases of sparse modality distribution or imbalanced co-occurrence, the hypergraph may become fragmented or biased toward dominant combinations, reducing feature aggregation effectiveness and limiting generalization to rare or atypical modality configurations.

2) Computational Complexity: The construction and maintenance of heterogeneous hypergraphs across all possible modality permutations incur considerable computational costs. This may limit the scalability of the framework, particularly for large-scale or real-time applications.

These challenges suggest promising directions for future research, including efficient hypergraph sparsification, uncertainty-aware fusion strategies, and adaptive sampling mechanisms to enhance modality distribution balance and scalability.

V. CONCLUSION

In this article, we propose a novel MHHL framework to address the challenges of incomplete multimodal semantic segmentation in remote sensing. Incomplete modality inputs often induce feature shifts and inconsistent representations across different modality combinations, severely impacting segmentation accuracy and robustness. To tackle these issues, MHHL introduces a heterogeneous hypergraph model that effectively captures the complex relationships among modalities under incomplete conditions. By leveraging the hypergraph structure, the model dynamically updates supernode features, enhancing its ability to model diverse land cover distributions. Furthermore, to mitigate feature shift, we propose a global multimodal feature modeling strategy combined with a multimodal hybrid feature discriminator. This design promotes better approximation and alignment of feature distributions across modality combinations, leading to more reliable and accurate segmentation outcomes. Extensive experiments on remote sensing datasets validate that MHHL outperforms existing methods in terms of both accuracy and robustness under incomplete modality settings.

Although the MHHL framework performs well in handling incomplete multimodal data, its representational capacity may be limited when the modality distribution is sparse or imbalanced. Future research will explore adaptive hypergraph construction to improve the framework's generalization in modality-incomplete and label-scarce scenarios.

APPENDIX A
APPENDIX THEORETICAL JUSTIFICATION OF HYPERGRAPH-BASED FEATURE MAPPING

This appendix provides a theoretical justification for the hypergraph-based feature mapping and updating mechanism employed in the proposed framework. The aim is to demonstrate that the design is not only empirically effective but also theoretically grounded in addressing the challenges of incomplete multimodal feature integration.

A. Formalization of the Hypergraph Structure

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ denote the constructed heterogeneous hypergraph, where $\mathcal{V}$ is the set of hypernodes, each representing a unique modality combination, and $\mathcal{E}$ is the set of hyperedges connecting hypernodes that share at least one common modality. Given n hypernodes, the input feature matrix at the tth graph convolutional layer is denoted by $\mathbf{H}^{(t)} \in \mathbb{R}^{n\times d'}$, where d' is the feature dimensionality. Each hypernode aggregates modality-specific features and serves as a functional unit for high-order multimodal representation learning. This formulation enables the model to flexibly encode modality-wise groupings in the presence of incomplete data.

B. Spectral Propagation and High-Order Modeling

The update mechanism in the proposed framework follows a normalized graph convolutional formulation, which approximates spectral filtering on a hypergraph:

$$\mathbf{H}^{(t+1)} = \sigma\left(\tilde{\boldsymbol{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\boldsymbol{D}}^{-1/2}\mathbf{H}^{(t)}\mathbf{W}^{(t)}\right) \tag{19}$$

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where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ is the adjacency matrix with added self-loops, $\tilde{\boldsymbol{D}}$ is its corresponding degree matrix, $\mathbf{W}^{(t)}$ denotes the trainable weight matrix, and $\sigma(\cdot)$ is a nonlinear activation function (e.g., ReLU). This update rule is equivalent to a first-order approximation of the graph Laplacian operator $g_\theta(\mathbf{L})$, with $\mathbf{L} = \mathbf{I} - \tilde{\boldsymbol{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\boldsymbol{D}}^{-1/2}$.

In this formulation, each propagation step acts as a low-pass filter in the spectral domain, smoothing feature signals over the hypergraph. This mitigates high-frequency noise and enhances the stability of learned representations, which is crucial when some modalities are incomplete. The smoothing effect serves as a contextual regularizer that enforces coherence among structurally related hypernodes.

Traditional graphs capture only pairwise relationships via binary edges. In contrast, hypergraphs allow modeling of higher order dependencies by connecting multiple nodes through a single hyperedge. This is particularly important in multimodal contexts where a sample may only contain a subset of available modalities.

Let $\mathbf{H} \in \mathbb{R}^{n\times d'}$ denote the hypernode feature matrix and $\mathbf{B} \in \{0,1\}^{n\times m}$ the incidence matrix of the hypergraph $\mathcal{G}$, where m is the number of hyperedges. The corresponding hypergraph Laplacian $\mathbf{L}_H$ is defined as

$$\mathbf{L}_H = \mathbf{I} - \boldsymbol{D}_v^{-1/2}\mathbf{B}\mathbf{W}_e\boldsymbol{D}_e^{-1}\mathbf{B}^\top\boldsymbol{D}_v^{-1/2} \tag{20}$$

where $\boldsymbol{D}_v$ and $\boldsymbol{D}_e$ are diagonal matrices representing node and hyperedge degrees, and $\mathbf{W}_e$ is a diagonal matrix of hyperedge weights.

The hypergraph Laplacian generalizes the notion of spectral propagation from simple graphs to high-order structures. It enables message passing among all nodes within a hyperedge, promoting semantically meaningful alignment across modality combinations. This flexibility provides a theoretical foundation for learning robust and generalizable representations under incomplete multimodal configurations.

C. Mutual Information Enhancement

The graph-based update rule also enhances mutual information between features from different modality combinations. Let $\mathbf{h}_i^{(t)}$ denote the feature vector of hypernode i at iteration t, and let $\mathcal{N}(i)$ represent its neighbor set. The update rule is given by

$$\mathbf{h}_i^{(t+1)} = \mathrm{ReLU}\left(\sum_{j\in\mathcal{N}(i)} \mathbf{W}^{(t)}\mathbf{h}_j^{(t)} + \mathbf{b}^{(t)}\right) \tag{21}$$

which implicitly increases the mutual information $\mathcal{I}(\mathbf{h}_i^{(t+1)}; \{\mathbf{h}_j^{(t)} | j \in \mathcal{N}(i)\})$. This is due to the aggregation of neighboring representations, which allows each node to incorporate semantic knowledge beyond its own modality configuration. Moreover, since both the linear transformation and activation function are Lipschitz continuous, the conditional entropy can be bounded as

$$\mathcal{H}\left(\mathbf{h}_i^{(t+1)}\right) \le \mathcal{H}\left(\left\{\mathbf{h}_j^{(t)}\right\}\right) + c \tag{22}$$

where c is a constant determined by the Lipschitz constants of the operations. This indicates that uncertainty in the node representation is progressively reduced, leading to more compact and semantically enriched features across multimodal groupings.

D. Robustness Under Incomplete Modalities

A key advantage of the hypergraph structure is its inherent ability to preserve semantic consistency and structural connectivity under conditions of missing modalities. Let $\mathcal{S}_i \subseteq \{1,2,3\}$ denote the set of available modalities for sample i. If two samples i and j satisfy $\mathcal{S}_i \cap \mathcal{S}_j \neq \emptyset$, then their corresponding hypernodes v_i and v_j are connected through shared hyperedges. This structural design ensures that message passing remains within semantically coherent neighborhoods, even in the absence of complete modality configurations.

Let $\mathbf{h}_i^{(T)}$ denote the feature of hypernode i after T iterations. The similarity between partially aligned modality sets is theoretically bounded by

$$\mathbb{E}_{\mathcal{S}_i\cap\mathcal{S}_j\neq\emptyset}\left[\left\|\mathbf{h}_i^{(T)} - \mathbf{h}_j^{(T)}\right\|^2\right] \le \epsilon \tag{23}$$

where ϵ is a small constant depending on the strength of the hyperedge weights and the spectral smoothness of the hypergraph. This ensures that representations of samples with shared modality subsets remain close in feature space, thereby enhancing robustness and generalization. Compared to modality-specific training or pairwise fusion schemes, the hypergraph-based formulation offers a principled mechanism to propagate complementary knowledge across samples with varying modality availability.

REFERENCES

[1] L. Zhang, Z. Tan, G. Zhang, W. Zhang, and Z. Li, "Learn more and learn usefully: Truncation compensation network for semantic segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4403814.

[2] Z. Chen et al., "SECBNet: Semantic segmentation-enhanced color balance network for optical satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4200313.

[3] B. Zhang, Y. Chen, W. Dang, S. Xiong, and X. Lu, "A spatial and semantic alignment fusion network for SeaLand port segmentation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 7420–7435, 2025.

[4] D. Hong et al., "SpectralGPT: Spectral remote sensing foundation model," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 8, pp. 5227–5244, Aug. 2024.

[5] Y. He, F. Huang, D. Wang, R. Zhang, X. Gu, and J. Pan, "NOMA-enhanced cooperative relaying systems in drone-enabled IoV: Capacity analysis and height optimization," *IEEE Trans. Veh. Technol.*, vol. 73, no. 12, pp. 19065–19079, Dec. 2024.

[6] L. Gao, J. Li, K. Zheng, and X. Jia, "Enhanced autoencoders with attention-embedded degradation learning for unsupervised hyperspectral image super-resolution," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, Apr. 2023, Art. no. 5509417.

[7] W. Li, J. Wang, Y. Gao, M. Zhang, R. Tao, and B. Zhang, "Graph-feature-enhanced selective assignment network for hyperspectral and multispectral data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5526914.

[8] J. Nie, C. Wang, S. Yu, J. Shi, X. Lv, and Z. Wei, "MIGN: Multiscale image generation network for remote sensing image semantic segmentation," *IEEE Trans. Multimedia*, vol. 25, pp. 5601–5613, 2023.

[9] G. Cheng, C. Yang, X. Yao, L. Guo, and J. Han, "When deep learning meets metric learning: Remote sensing image scene classification via learning discriminative CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2811–2821, May 2018.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 10:05:32 UTC from IEEE Xplore. Restrictions apply.

[10] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.
[11] J. Li, Y. Liu, R. Song, Y. Li, K. Han, and Q. Du, "Sal^{2}RN: A spatial–spectral salient reinforcement network for hyperspectral and LiDAR data fusion classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5500114.
[12] X. Li et al., "MCANet: A joint semantic segmentation framework of optical and SAR images for land use classification," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 106, Feb. 2022, Art. no. 102638.
[13] M. K. Reza, A. Prater-Bennette, and M. S. Asif, "MMSFormer: Multimodal transformer for material and semantic segmentation," *IEEE Open J. Signal Process.*, vol. 5, pp. 599–610, 2024.
[14] H. Liu, W. Li, X.-G. Xia, M. Zhang, Z. Guo, and L. Song, "SegHSI: Semantic segmentation of hyperspectral images with limited labeled pixels," *IEEE Trans. Image Process.*, vol. 33, pp. 6469–6482, 2024.
[15] X. He, Y. Zhou, J. Zhao, D. Zhang, R. Yao, and Y. Xue, "Swin transformer embedding UNet for remote sensing image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4408715.
[16] W. Jiang, K. Huang, J. Geng, and X. Deng, "Multi-scale metric learning for few-shot learning," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 31, no. 3, pp. 1091–1102, Mar. 2021.
[17] Y. Chen, Y. Wang, S. Xiong, X. Lu, X. X. Zhu, and L. Mou, "Integrating detailed features and global contexts for semantic segmentation in ultrahigh-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4703914.
[18] Z. Xu, W. Jiang, and J. Geng, "Dual-branch dynamic modulation network for hyperspectral and LiDAR data classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514813.
[19] J. Zhang, H. Liu, K. Yang, X. Hu, R. Liu, and R. Stiefelhagen, "CMX: Cross-modal fusion for RGB-X semantic segmentation with transformers," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 12, pp. 14679–14694, Dec. 2023.
[20] B. Yin, X. Zhang, Z. Li, L. Liu, M.-M. Cheng, and Q. Hou, "DFormer: Rethinking RGBD representation learning for semantic segmentation," 2023, *arXiv:2309.09668*.
[21] A. Sharma and G. Hamarneh, "Missing MRI pulse sequence synthesis using multi-modal generative adversarial network," *IEEE Trans. Med. Imag.*, vol. 39, no. 4, pp. 1170–1183, Apr. 2020.
[22] Y. Shen and M. Gao, "Brain tumor segmentation on MRI with missing modalities," in *Proc. Int. Conf. Inf. Process. Med. Imag.*Cham, Switzerland: Springer, 2019, pp. 417–428.
[23] Y. Wang, Z. Cui, and Y. Li, "Distribution-consistent modal recovering for incomplete multimodal learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 21968–21977.
[24] Y.-L. Lee, Y.-H. Tsai, W.-C. Chiu, and C.-Y. Lee, "Multimodal prompting with missing modalities for visual recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 14943–14952.
[25] M. Ma, J. Ren, L. Zhao, S. Tulyakov, C. Wu, and X. Peng, "SMIL: Multimodal learning with severely missing modality," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2302–2310.
[26] H. Maheshwari, Y.-C. Liu, and Z. Kira, "Missing modality robustness in semi-supervised multi-modal semantic segmentation," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2024, pp. 1009–1019.
[27] S. Wei, C. Luo, and Y. Luo, "MMANet: Margin-aware distillation and modality-aware regularization for incomplete multimodal learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 20039–20049.
[28] J. Nie, X. Wang, R. Hou, G. Li, H. Chen, and W. Zhu, "Dynamic spatio-temporal graph reasoning for VideoQA with self-supervised event recognition," *IEEE Trans. Image Process.*, vol. 33, pp. 4145–4158, 2024.
[29] C. Zheng et al., "Cross-modal progressive perspective matching network for remote sensing image-text retrieval," *IEEE Trans. Multimedia*, vol. 27, pp. 3966–3978, 2025.
[30] E. Xie et al., "SegFormer: Simple and efficient design for semantic segmentation with transformers," in *Proc. Adv. Neural Inf. Process. Sys. (NIPS)*, vol. 34, Dec. 2021, pp. 12077–12090.
[31] M. Liu, Y. Gao, P.-T. Yap, and D. Shen, "Multi-hypergraph learning for incomplete multimodality data," *IEEE J. Biomed. Health Informat.*, vol. 22, no. 4, pp. 1197–1208, Jul. 2018.
[32] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *J. Roy. Stat. Soc. B, Stat. Methodol.*, vol. 39, no. 1, pp. 1–22, Sep. 1977.
[33] Y. Li, Y. Zhou, Y. Zhang, L. Zhong, J. Wang, and J. Chen, "DKDFN: Domain knowledge-guided deep collaborative fusion network for multimodal unitemporal remote sensing land cover classification," *ISPRS J. Photogramm. Remote Sens.*, vol. 186, pp. 170–189, Apr. 2022.
[34] J. Zhang et al., "Delivering arbitrary-modal semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 1136–1147.
[35] X. Ma, X. Zhang, M.-O. Pun, and M. Liu, "A multilevel multimodal fusion transformer for remote sensing semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5403215.
[36] X. Ma, X. Zhang, M.-O. Pun, and B. Huang, "MANet: Fine-tuning segment anything model for multimodal remote sensing semantic segmentation," 2024, *arXiv:2410.11160*.



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